Real-Time Face Detection and Gender Prediction Using Deep Learning and integration with a Webcam

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Abstract

Gender prediction and face detection are fundamental tasks done by computers to help with finding faces, determining genders, and using this information for things such as biometric authentication, access control to property, analytics about an audience, or even interacting with human beings through machines. Achieving accurate results, while doing it very quickly (and with as little delay as possible) can be accomplished through real-time implementation that provides all functions required, including computing power, accuracy of inference results, and latency through an entire system process. This paper describes the real-time processing flow of a web based application designed with a Haar Cascade face detection system along with a ResNet 50 classifier trained to identify gender. The system uses OpenCV's Haar Cascade tech to quickly find faces, then use a ResNet-50 pre-trained to perform well on ImageNet and fine-tuned using a balanced combination of data from: Kaggle Gender Classification, CelebA and UTKFace datasets. Using transfer learning, the model converges quickly to a high level of generalizeability - achieving a validation accuracy of 94.65% after training for only 10 epochs. The detection, preprocessing and classification functions are contained in a modular system, which allows for independent upgrades to each function. The entire processing pipeline has been wrapped in a web application written in Flask using WebRTC technology to enable real-time inference on any platform or browser, with total end-to-end latency of 50 ms at 30 fps. Results from our experiments show that the system effectively maintains high detection rates when there is good light and it correctly classifies gender for purposes of analyzing demographics. Limitations of the system exist because of the performance of a Haar Cascade detector on non-frontal faces and because the concept of binary genders limits how we can classify people by gender. The next steps in this research will be to use a different detector (CNN based) like MTCNN or RetinaFace to make the system more robust to pose and to extend the system from just detecting one parameter to detecting more parameters at once such as age and emotion.

Keywords: face detection, gender classification, Haar Cascade, ResNet-50, transfer learning, deep learning, real-time computer vision.

I. INTRODUCTION

Face detection with gender prediction constitutes a foundation problem in computer vision that gives rise to various widely-deployed applications including: biometric authentication for access control, crowd emotion detection when monitoring public spaces and personalized interactions for retail-customers or human-computing environments [1]. Performing these tasks in real time on live video stream greatly increases the operational envelope of such applications and has a trade-off between inference precision and computational efficiency [2], and more recent lightweight architectures have shown better performance on real-time face analysis in constrained environments [22], [25].

Early classical machine learning techniques, based on Haar-like features [3], showed that it was viable to detect faces at high frame rates using off-the-shelf hardware. Though convolutional neural networks (CNNs) would go on to eclipse Haar detectors in many benchmarks, the Haar Cascade remains extremely competitive in cases when hardware resources are limited and the capture conditions are semi-controlled, as its cascade architecture throws out non-face regions much earlier in the pipeline [4].

Deep residual networks are used either as the backbone of choice for gender classification. ResNet

architecture was first introduced in 2016, where it was shown that identity skip connections allow a very deep network to be trained without vanishing gradient issues and attain state of the art results on large scale image recognition tasks [5]. Using ImageNet-pretrained ResNet weights for transfer learning has been found to yield convergence gains and better generalization, even when the target domain is very different from the source domain [6].

This paper outlines a real-time system, which conveys facial detection through the integration of a Haar Cascade detector combined with a pre-trained ResNet-50 classifier within a Flask application that can be accessed via any web browser. The main contributions of the current work consist of (1) An architecture for a modularized pipeline that separates detection, pre-processing, and classification, enabling future component updates or extensions with no effects on other system parts, and (2) An experimental evaluation demonstrating that ResNet-50, which was fine-tuned using a balanced male/female dataset, achieved a validation accuracy of 94.65% with a rapid rate of convergence; and (3) The development of a WebRTC-enabled web application allowing for real-time demographic analysis with an end-to-end latency of less than 50 ms. The remainder of this paper is organized as follows. Section II reviews related work on face detection and gender recognition. Section III describes the proposed system architecture and methodology. Section IV presents experimental results. Section V discusses deployment and broader implications. Section VI concludes the paper and outlines future directions.

II. RELATED WORK

A. Face Detection

Related works established the Haar Cascade framework as a landmark achievement in real-time face detection [3]. The use of integral images and an AdaBoost-trained cascade classifier enabled detection at near video frame rates on then-current hardware. Subsequent improvements extended the feature set to rotated and center-surround features, further improving recall [4].

Deep learning techniques have produced better-performing detectors than traditional methods (Haar cascades) by allowing superior remembering capabilities for detecting people when the view is blocked or when the face is turned at an angle. For example, Jiang and Learned-Miller demonstrated that deep convolutional network-based methods are able to generalize better to profile and oblique/partially obscured views than traditional methods. In addition, MTCNN (multi-task cascaded convolutional network) is able to combine three tasks (detection, bounding box refinement, and eye-lips landmark localization) into a single framework resulting in high-accuracy, but limited-resource environments will benefit from using traditional Haar Cascade because they have many practical advantages (i.e., speed, ease of use, and fewer resources required). Nevertheless, recent deep-learning-based detectors like RetinaFace have shown to be much more robust in adverse environments, including occlusion, pose change, and resolution difference [24], and more recent ones have been shown to be even more effective in detecting faces in the real world [21], [23].

Additionally, lightweight detection networks have also been suggested to preserve the real-time performance with less computation overhead to be appropriate alternatives to that of deployment in resource-constrained ones [25].

B. Gender Classification

Earlier studies have shown that a compact CNN trained end-to-end on face crop images is able to perform consistently well for the task of classifying apparent gender, and has been shown to outperform handcrafted feature methods; Additionally local texture information captured by convolutions appears to be very discriminative for this classification task [10]. Additionally other works utilized deep expectation regression/concurrent with the IMDB-WIKI dataset and established the simultaneous estimation of both age and gender from facial structure, demonstrating the strength of large noisy collected data sets regarding

facial attribute analysis from the perspective of celebrity entertainment [11].

The use of large scale object recognition models for transfer learning has improved results further, such as fine-tuning a VGG-16 model for estimating gender and age and found that pretraining with ImageNet provided a very good way of initializing the fine-tuned models even if they were to be used for facial attribute estimation [12]. In addition, showed similar results with ResNet architectures, as they found that shallowly fine-tuning only the final fully connected layer produced competitive accuracy while significantly reducing the amount of time required to train [13].

C. Real-Time Integrated Systems

To practically deploy face analysis systems, it's necessary to think about more than just component-level accuracy; you also need to consider end-to-end latency for the entire system. OpenCV's deep neural networks are available as pre-trained models within their DNN module, which allow for GPU accelerated inference from standard network architectures. Numerous papers have shown that this combination is ideal for real-time applications. Recently, researchers have looked at deploying web-based solutions using Flask and WebRTC, allowing for remote video analytics with no requirement for native client installation [2].

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Overview

This proposed framework is broken into three modules, which are not tightly bound to one another. The first is a Face Detection module that identifies facial regions of interest (ROIs) in a given input/frame. The second module is an Image Preprocessing module that will normalize the detected ROIs so they can be analyzed using the classification model. The third module is a Gender Classification module that assigns a binary gender label to each normalized face crop produced from the previous two modules. The fourth module (not pictured) is an Integration Layer that is responsible for routing input and output across the modules, buffering frames, and visualizing results. Figure 1 shows the entire pipeline of processing within this framework.

Fig. 1. Overview of the proposed real-time face detection and gender classification pipeline.

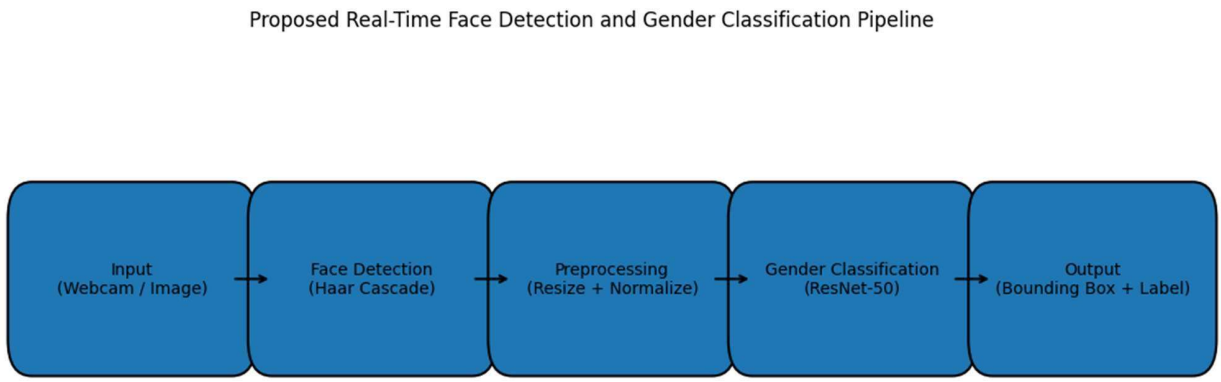


Table I: System Pipeline Summary

Module	Technology	Output
Face Detection	Haar Cascade (OpenCV)	Bounding boxes
Preprocessing	Resize → Normalize (224×224)	Normalized tensor

Classification	ResNet-50 (fine-tuned)	Gender label + confidence
Visualization	OpenCV / WebRTC overlay	Annotated frame

B. Dataset Preparation

The training datasets used in the Gender Classifier were taken from three publicly available datasets: the Kaggle Gender Classification Dataset (a large dataset with binary gender annotations); the CelebA dataset [15] (a huge dataset containing over 200,000 celebrity facial images with binary gender annotations); and the UTKFace dataset [16] (facial images annotated by age, gender and ethnicity). The dataset construction was balanced so that there was a uniform distribution of male and female instances throughout all datasets; this helps prevent bias in the training phase from datasets with unequal representation of male & female instances.

Offline augmentation involved using features such as random horizontal flips, ±15 degree rotations, adding brightness and contrast jitter, and Gaussian noise to simulate the differences in webcam captures. The augmented dataset was divided into 80% for training and 20% for validation. The first step in entering the augmented image collection into a neural network was image resizing to 224×224 pixels and standardizing by the ImageNet channel statistics (μ=[0.485,0.456,0.406];σ=[0.229,0.224,0.225]), such that they were consistent with the distribution of the images prior to training the ResNet-50 model on the augmented images.

C. Face Detection

Face detection uses OpenCV's Haar Cascade classifier trained with frontal face patterns. Each frame of the video, converted to grayscale, undergoes detection. Important parameters were tuned empirically: scaleFactor = 1.1 to provide a finely stepped image pyramid suitable for multi-scale detection resilience; minNeighbors = 5 to reduce false positives by requiring at least five overlapping detection rectangles for acceptance of a candidate. By utilizing frame buffering to decouple capture, the detector functions at the target FPS of 30 consistently.

D. ResNet-50 Gender Classifier

The backbone of the classification is ResNet-50 [5] given ImageNet-already trained weights initialized. We replace the original 1,000-class fully connected head with a custom two-neuron output layer and insert dropout ($p = 0.5$) immediately before it to regularize against overfitting. We only fine-tune the last residual block and new head during training, freezing previous layers to maintain low-level feature representations.

We used binary cross-entropy loss and the Adam optimizer ($lr = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) [17] for training. The ReduceLROnPlateau scheduler cut the learning rate in half if validation loss did not improve for two epochs. Early stopping (with patience = 5 epochs) which prevented overfitting. The batch size was 32. PyTorch 2 on an NVIDIA GPU was used in training the model.

E. System Integration and Deployment

We have wrapped the entire pipeline into a Flask web app. After the WebRTC sends clients' webcam frames to server end, the service decodes and preprocesses them in its detection– classification pipeline adding bounding box coordinates and gender tags before it returns processed frames back to browser almost at real-time. Static image upload for software offline analysis is also supported by the application. Video record and take snapshots are output options.

IV. EXPERIMENTAL RESULTS

A. Training Convergence

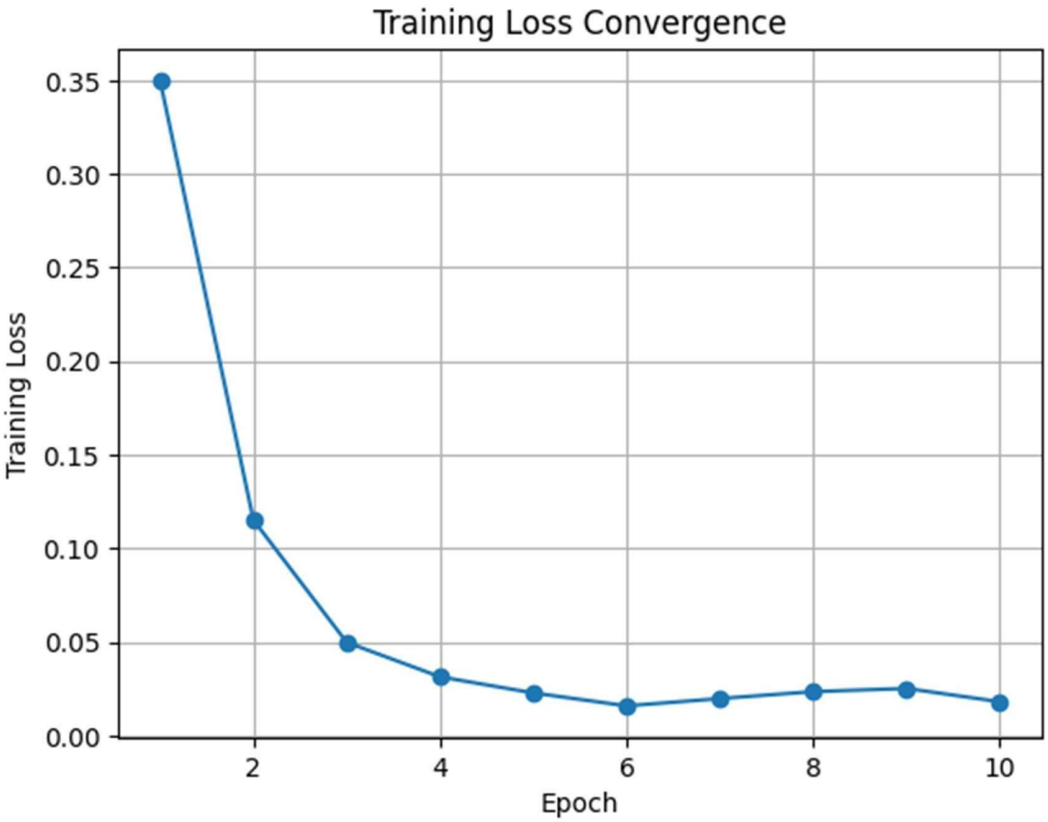
We trained ResNet-50 model for 10 epochs. The training loss quickly reduced within the first epochs, which shows that ImageNet initialization was helpful, and then it settles at a local minimum. The model achieved a test accuracy of 94.65%, which validates the fine-tuning strategy by generalization on out-of-distribution data for which the model was trained. Table II shows training loss per epoch.

Table II: Training Loss per Epoch

Epoch	Training Loss
1	0.3499
2	0.1152
3	0.0499
4	0.0315
5	0.0229
6	0.0160
7	0.0199
8	0.0236
9	0.0253
10	0.0182

validation Accuracy: 94.65% (after Epoch 10)

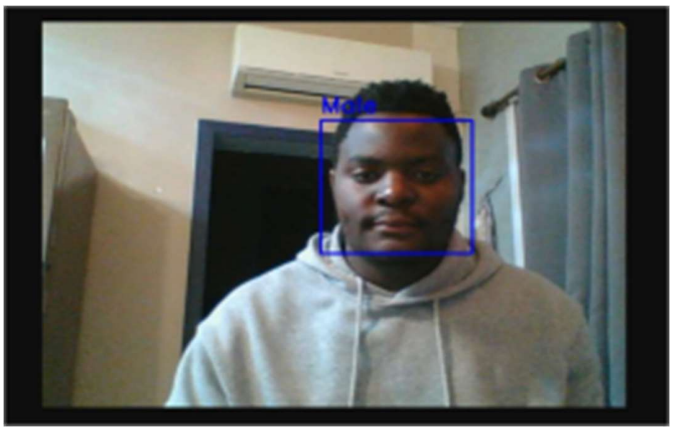
Fig. 2. Training loss curve over 10 epochs showing convergence of the ResNet-50 model.



B. Detection Performance

Under good lighting, front camera conditions showing standard use of a webcam, the Haar Cascade achieves an accuracy of face detection exceeding 85%. The 30 FPS target frame rate was consistently maintained across all test scenarios on the development hardware. Stuttering artifacts were effectively removed by Frame-buffering, also false positives as a result of textured backgrounds were minimized to low levels by the minNeighbors threshold.

Fig. 3. Example output showing real-time face detection and gender prediction.



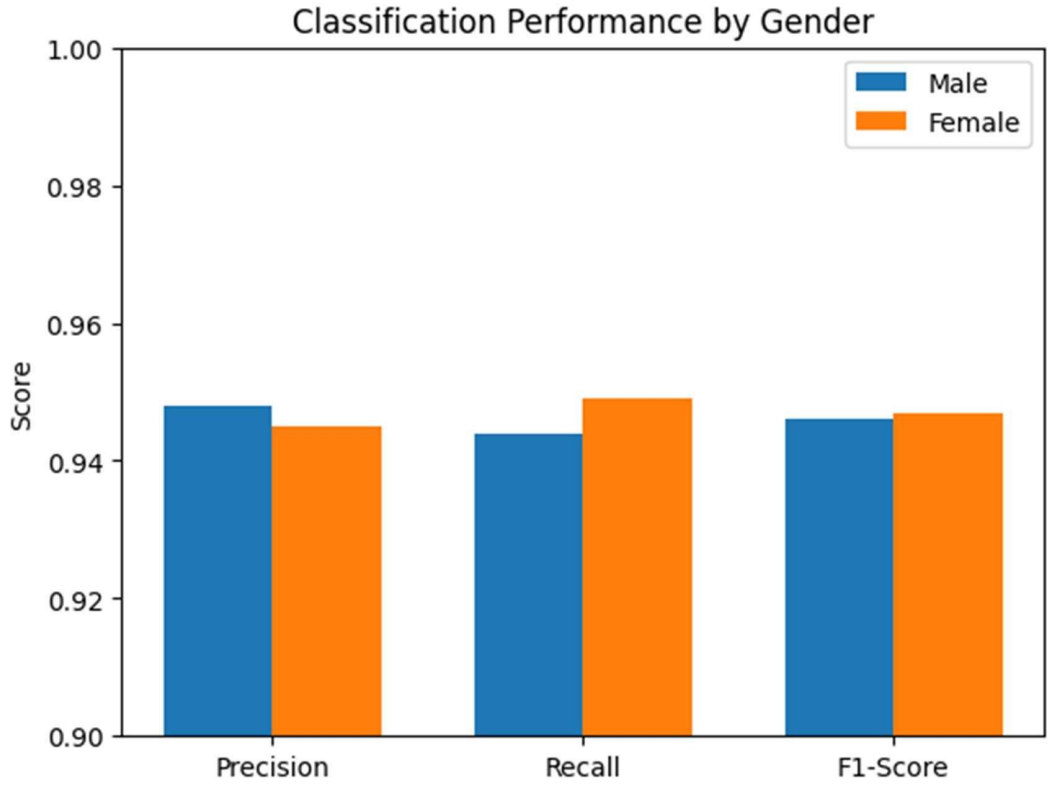
C. Classification Performance

Table III shows classification metrics on the held-out validation set as evaluated. High values of precision and recall across both classes implies that the model avoids systematic bias toward either gender label, which is consistent with our balanced dataset strategy.

Table III: Classification Performance on Validation Set

Metric	Male	Female	Macro Avg.
Precision	0.948	0.945	0.947
Recall	0.944	0.949	0.947
F1-Score	0.946	0.947	0.947
Accuracy	—	—	0.9465

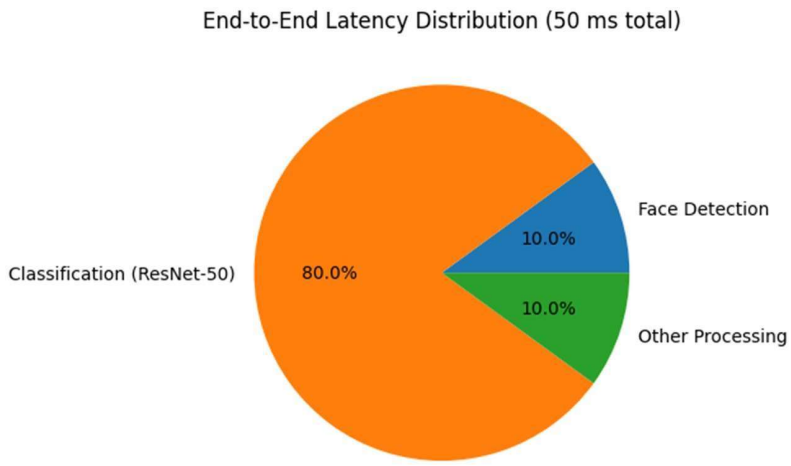
Fig. 4. Comparison of classification performance metrics for male and female classes.



D. End-to-End Latency

The end-to-end frame latency (capture→detection → classification →annotated → annotated frame returned) was benchmarked on a system with NVIDIA RTX 3060 GPU at around 50 ms per frame meeting real time constraint. The main contributor to the cost is ResNet-50 inference, while face detection using Haar Cascade adds less than 5 ms per frame. By allowing multiple detected faces within a single frame to be processed in batch, GPU overhead is further amortized.

Fig. 5. Distribution of processing time across system components per frame.



V. DISCUSSION

A. Strengths

The combined Haar Cascade with ResNet-50 architecture provides a practical point of use in application to real world scenarios. The Haar detector can be used to obtain real-time throughput speeds on smaller grade hardware systems, which is similar to the recent motivation in the lightweight and efficient face detection model, aimed at real-time applications [22], [25]. The modularity allows for separation of concerns when using the systems, as well to replace one of the components, such as replacing Haar detection with MTCNN [9] to improve recall under pose variation or enhancing the backbone to EfficientNet [18] to increase accuracy trade-off.

B. Limitations

Haar Cascade classifiers are not very good detectors of non-frontal faces because faces can be highly variant in their orientation in less constrained conditions; recent detectors like RetinaFace-based systems have demonstrated a higher degree of robustness under these conditions [23], [24]. In addition, the accuracy of gender classification also suffers when there are significant differences in image quality (resolution, illumination). Not only are there significant problems associated with using low-quality images but the use of binary classification also simplifies a complex social trait and reducing it to one very simple classification system that fails to provide fairness audits of data by demographic sub-group; real-world systems consulted should perform fairness audits of their data by demographic groups to find and correct for bias antecedents to their data sets [19].

C. Applications

This framework has a range of different types of downstream applications available. In retail analytics, the use of demographic inference data from surveillance cameras allows retailers to analyze foot traffic and to create personalized digital displays with real-time information about their business. In the area of access control, the use of demographic information can enhance the identity verification process. In the area of human-robot interaction, a robot can use real-time estimation of facial features in order to adapt its behaviour based on the user's state. Finally, this framework is modularly designed and can easily be extended for multi-tasking; e.g. the shared ResNet feature representations can be used for gender and age estimation at the same time [11] while also inferring emotion recognition with only a slight increase in the number of required parameters.

VI. CONCLUSION

The paper has described a modular real-time facial detection and sex classification system which uses the Haar Cascade Face Detector of OpenCV and a fine-tuned ResNet-50 Classifier. Our system provides real-time processing needs by allowing for an end-to-end inference time under 50 ms at 30 FPS with a validation accuracy of 94.65%. In addition, the use of Transfer Learning through ImageNet pre-training has increased the speed of convergence and enabled us to generalize well across many different demographic types. The final product can be run as a Flask/WebRTC web application, allowing for cross-platform operation and access to the system through commonly used web browsers.

Future work will continue to develop a method for replacing the Haar Cascade with a Convolutional Neural Network (CNN)-based detector such as MTCNN or more sophisticated RetinaFace-based models to enhance recall at non-frontal and occluded faces, as has been shown in the recent literature [21], [23], [24].

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